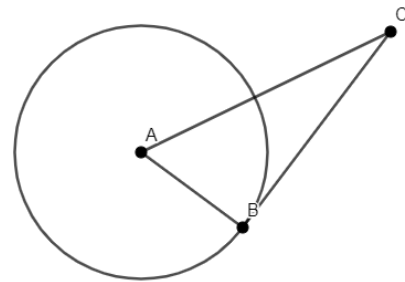
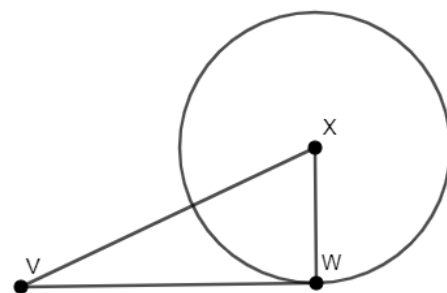


1. Circle A is shown with point C outside the circle, where point B is on the circle, $AC = 10$, $CB = 8$, and $BA = 6$. Verify if $\overline{BC}$ is tangent to circle A using the Pythagorean Theorem.



Circle X on the right has $\overline{VW}$ as a tangent segment where W is the point of tangency. $VX = 13$ and $VW = 12$.

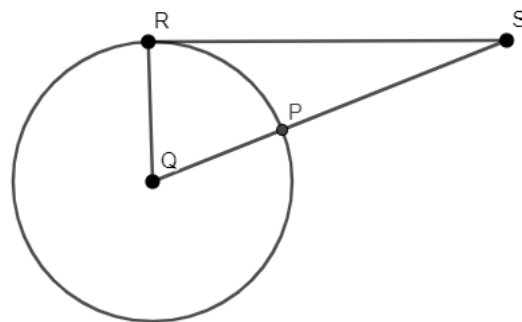
2. What is the length of $\overline{XW}$?



3. What is the area of $\triangle XWV$?

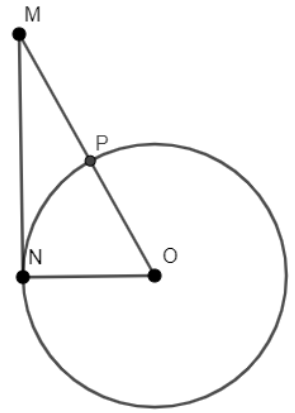
In Circle Q , $\overline{RS}$ is tangent to the circle at point R , $QR = 7$, and $PS = 18$.

4. What is the length of $\overline{RS}$?



5. What is the area of $\triangle QRS$?

In Circle O , $\overline{MN}$ is a tangent segment whose point of tangency is point N . $MN = 15$ and $MP = 6$.

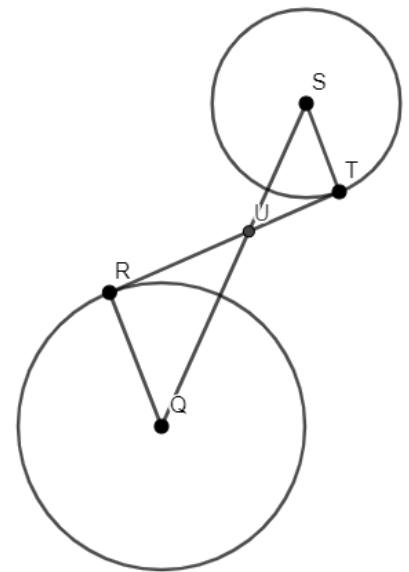


6. What is the length of $\overline{ON}$?

7. What is the length of $\overline{MO}$?

8. What is the perimeter of $\triangle MNO$?

In the figure on the right, $\overline{RT}$ is a common tangent segment of both Circle Q and Circle S . $QR = 15$, $RU = 8$, and $TU = 6$.



9. What is the length of $\overline{ST}$?

10. What is the length of $\overline{QS}$?